THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 12: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The diagonal of a square is $7\sqrt{2}$ cm. Its perimeter is: 1

(a) 35 cm **(b)** 36 cm **(c)** 42 cm **(d)** 28 cm

2. A rhombus has side 25 cm and one diagonal 14 cm. The other diagonal is: 1

(a) 50 cm **(b)** 48 cm **(c)** 46 cm **(d)** 72 cm

3. The diagonals of a rhombus are 42 cm and 144 cm. Its area is: 1

(a) 3168 cm^2 **(b)** 3024 cm^2 **(c)** 3066 cm^2 **(d)** 3034 cm^2

4. The diagonal of a rectangle is 123 cm and one side is 27 cm. Its area is: 1

(a) 3360 cm^2 **(b)** 3250 cm^2 **(c)** 3267 cm^2 **(d)** 3240 cm^2

5. In a parallelogram, two opposite angles are $(2x + 4)^\circ$ and $(x + 105)^\circ$. The value of x is: 1

(a) 100 **(b)** 102 **(c)** 101 **(d)** 103

6. The diagonals of a quadrilateral are 110 cm and 92 cm. The perimeter of the parallelogram formed by joining the midpoints of its sides is: 1

(a) 606 cm **(b)** 303 cm **(c)** 406 cm **(d)** 202 cm

7. The four angles of a quadrilateral are in AP with common difference 20°. The smallest angle is: 1

(a) 40° **(b)** 80° **(c)** 70° **(d)** 60°

8. The sides of a rectangle are in the ratio $3 : 4$ and its diagonal is 40 cm. Its perimeter is: 1

(a) 168 cm **(b)** 152 cm **(c)** 112 cm **(d)** 120 cm

9. **Assertion (A):** The diagonals of a rhombus are equal. 1

Reason (R): The diagonals of a rhombus bisect each other at right angles.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** Every rectangle is a parallelogram. 1

Reason (R): The diagonals of a rectangle are equal.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The angles of a quadrilateral are in AP with common difference 10°. Find all four angles and say whether a pair of them is supplementary. **2**

12. In parallelogram ABCD, $AB = (2x + 5)$ cm, $CD = (x + 21)$ cm and $BC = (x + 4)$ cm. Find x, the sides and the perimeter. **2**

13. A rhombus has side 75 cm and one diagonal 42 cm. Find the other diagonal, its area and the distance between a pair of opposite sides (height), correct to two decimal places. **2**

14. The angles of a quadrilateral are in AP with common difference 14°. Find all four angles and say whether a pair of them is supplementary. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The diagonals of a parallelogram ABCD meet at O. $OA = (5x + 5)$ cm and $OC = (x + 77)$ cm. Find x and the length of AC. **3**

16. In rhombus ABCD, $\angle A = 78°$. Find $\angle B$, and the angles $\angle OAB$ and $\angle OBA$ where O is the point of intersection of the diagonals. **3**

17. In $\triangle ABC$, D is the midpoint of AB and the line through D parallel to BC meets AC at E. $BC = 18$ cm and $AB = 28$ cm. Find DE, AD and show that E is the midpoint of AC. **3**

OR

The diagonals of a parallelogram ABCD meet at O. $OA = (5x + 5)$ cm and $OC = (3x + 59)$ cm. Find x and the length of AC. **3**

18. In rhombus ABCD, $\angle A = 138°$. Find $\angle B$, and the angles $\angle OAB$ and $\angle OBA$ where O is the point of intersection of the diagonals. **3**

OR

In $\triangle ABC$, D is the midpoint of AB and the line through D parallel to BC meets AC at E. $BC = 40$ cm and $AB = 24$ cm. Find DE, AD and show that E is the midpoint of AC. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A tile in the shape of a parallelogram PQRS is used in a floor design. $\angle P = 65°$, $PQ = 8$ cm, $QR = 5$ cm. Its diagonals PR and QS meet at O, with $PR = 14$ cm and $QS = 10$ cm. **5**

(i) Find $\angle R$.

(ii) Find $\angle Q$.

(iii) Find the perimeter of the tile.

(iv) Find OP and OQ and state the property used.

20. The midpoints of the sides of a rectangle 30 cm by 72 cm are joined in order. Show that the figure formed is a rhombus and find its perimeter and area. **5**

OR

In a parallelogram ABCD the diagonals meet at O with $OA = 24$ cm, $OB = 45$ cm and $AB = 51$ cm. Show that ABCD is a rhombus and find its perimeter. **5**

END OF QUESTION PAPER — Total: 40 Marks